

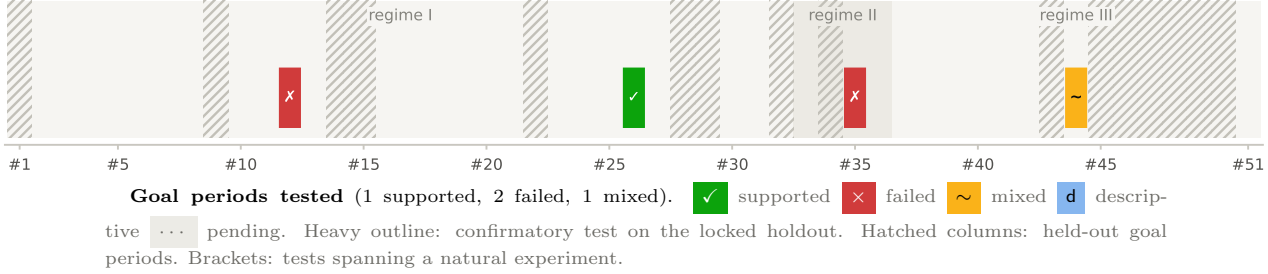
H115: Blind role recovery in the debate week: the judge is a sink of antisymmetric coupling (#12)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 1.5/5

Bottom line. The judge is not recoverable blind as a talk sink (rank 1 in 3/10 debates; a sink ≥ 0.6 is excluded), and debate talk couples across teams, not within them ($J_{ST} - J_{OT} = -0.16 \pm 0.08$, $p = 0.01$).

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

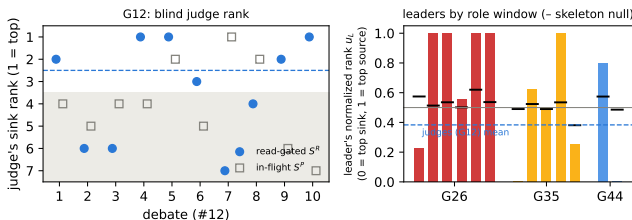
HH353 asked whether the per-call talk kinetic Ising, fitted blind to timing, ranks each debate's judge as the top sink of antisymmetric coupling. In the ten #12 debates (7 agents, 3,644 calls) the ranking was frozen and hashed before any role label was read. The judge ranks first in 3 of 10 debates (median rank 2.5, mean normalized rank 0.38; uniform $p = 0.15$, field-only skeleton $p = 0.24$). The predicted team blocks come out reversed: debaters talk after reading an opponent, not a team-mate. As replication, the #26 elected leader is the top talk source in 4 of 6 windows; #35 lead designers and the #44 leader show nothing. The holdout is not run.

PHYSICAL MODEL

Model 02. Spin $s_i(c) = \pm 1$: call c of agent i talks or not, on the ledger call clock. Input $x_{ij}(c) = 1$ if j posted in i 's room since i 's previous call (read-gated); x_{ij}^P : posted during the call (in-flight placebo). Fields h : agent $\times$ call mode and agent $\times$ debate phase. Couplings $J_{ij} = \alpha_i + \beta_j$ (in, out); sink score $S_i = \text{in} - \text{out}$. Pair-type model: J by same team, opposite team, judge and bench.

$$P(s_i(c) = +1) = \frac{e^{H_i(c)}}{2 \cosh H_i(c)}, \quad H_i = h_{i,\kappa} + h_{i,\phi} + J_{ii}s_i(c^-) + \sum_j J_{ij}x_{ij} + \sum_j J_{ij}^P x_{ij}^P$$

$$S_i = \sum_j (J_{ij} - J_{ji}), \quad u = \frac{\text{rank}(S) - 1}{n - 1} \quad (0 = \text{top sink})$$



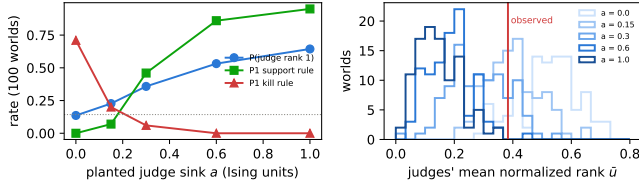
Observables. Left: the judge's blind sink rank in each debate under read-gated (blue) and in-flight (grey) couplings; the shaded band is the HH kill region. Right: the designated leader's normalized rank per role window (bars) against its field-only skeleton null (black ticks); 1 is the top source. The #26 leader is a source; judges lean slightly toward the sink end.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a monitor: one-way roles do not show up as one-way talk timing at debate scale. A judge whose talk follows debaters' talk by ≥ 0.6 Ising units per read debater is excluded, so blind role detection from who-talks-after-whom fails for 7 agents and 300 calls per episode. Opposition does show up: opponents answer each other (cross-team J 0.20 vs within-team 0.04). A cheap opposing-camp detector is therefore cross-block talk coupling, not within-block coupling. An elected leader can be the top talk source, but a single-agent rank needs a skeleton null (its null mean is 0.38 to 0.62, not 0.5).

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 judge top sink median 2.5, rank 1 in 3/10; $\bar{u} = 0.38$ (median ≤ 2 , rank 1 in ≥ 4)		not supported
P1b within-agent contrast > 0	+0.39 (perm. p 0.12)	not supported
P1c read beats in-flight	mean rank 3.3 vs 4.0	direction only
P2 $J_{ST} - J_{OT} > 0$	-0.16 ± 0.08 ; perm. $p(\leq) = 0.01$	failed (reversed)
P3 rulings as steps	$J_{DJ} - J_{DJ}^P = +0.02 \pm 0.11$; $ \Delta h $ in band	failed
R1 leaders in half ($\geq 2/3$)	9/13 pooled; G26 5/6, G35 3/5, G44 1/2	G26 only
Synthetic (F)	size ≤ 0.01 ; 80% power at sink $a = 0.6$; team power 0.87	passed



Second figure. Synthetic validation on the real #12 skeleton and call order (100 worlds per point). Left: judge rank-1 rate, support rule and kill rule against the planted sink a . Right: the judges' mean normalized rank in planted worlds; the observed value (red) sits at the median of $a = 0.15$ worlds and excludes $a = 0.6$.

GOAL PERIODS IN DETAIL

G12 $\times$ failed
G26 $\checkmark$ supported
G35 $\times$ failed
G44 $\sim$ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Regime-I talk may not run on the all-call clock (H67); the chat-mode clock variant moves the judge to median rank 4 but has little power. Per-debate fits use 170–630 calls, so only large role signals are resolvable. The pooled judge in- vs out-coupling ($J_{JD} = 0.28$, $J_{DJ} = 0.07$) points the predicted way but uses labels and is post hoc. The leader-as-source replication rests on one unit (Fisher $p \approx 0.08$). The verdict step's sign reversal was found after the pre-registered $|\Delta h|$ rule failed.

ROUND 2

Blind team recovery from cross-block coupling (antiferromagnetic ground state); split reads into named and unnamed messages; confirm the focal-agent source rank on NE37 (#48).

- **H115-R1** Blind team recovery from cross-team coupling.
- **H115-R2** Addressed reads.
- **H115-R3** Leader as source, more units.
- **H115-R4** Regime-I clock.